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Project Title:
Hospital Readmission Prediction Using Machine Learning

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Declaration

I hereby declare that this project titled **“Hospital Readmission Prediction Using Machine Learning”** is my original work carried out under the guidance of the academic team of Denvey EduGrow. This project has not been submitted elsewhere for any academic degree or certification.

Student Signature
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Date 7/1/26

Acknowledgement

I would like to express my sincere gratitude to **Denvey EduGrow** for providing the opportunity and guidance to complete this project. I also thank the mentors and instructors for their continuous support and valuable feedback throughout the project duration.

Abstract

Hospital readmission is a major challenge in the healthcare system as it increases treatment costs and reflects the quality of patient care. Predicting whether a patient will be readmitted within a short period after discharge can help hospitals take preventive measures. This project focuses on building a machine learning model to predict hospital readmission using patient medical and demographic data. The dataset is preprocessed and relevant features are selected for training the model. Logistic Regression is used as the primary classification algorithm to predict readmission. The results show that machine learning techniques can effectively assist healthcare professionals in reducing hospital readmission rates.

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1. Introduction

Hospital readmission occurs when a patient is admitted to a hospital again within a short time after being discharged. High readmission rates indicate poor healthcare management and increase the financial burden on hospitals and patients. With the growth of healthcare data, machine learning techniques can be used to analyze patient records and predict the risk of readmission. This project aims to develop a predictive model that helps identify high-risk patients in advance.

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2. Problem Statement

Hospitals face challenges in identifying patients who are likely to be readmitted after discharge. Manual prediction is difficult due to the large amount of patient data. There is a need for an automated system that can accurately predict hospital readmission using machine learning techniques.

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3. Objectives

1. To study hospital readmission and its impact on healthcare

2. To analyze patient medical and demographic data

3. To preprocess and clean the dataset

4. To build a machine learning model for prediction

5. To evaluate the performance of the model

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4. Dataset Description

The dataset used in this project is a publicly available, de-identified hospital dataset containing patient medical and demographic information and does not include any personally identifiable information. It includes attributes such as age, number of lab procedures, number of medications, length of hospital stay, and previous admissions. The target variable indicates whether the patient was readmitted or not. The dataset is preprocessed to handle missing values and categorical variables.

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5. Methodology

The methodology followed in this project includes:

1. Data collection from a public source

2. Data cleaning and preprocessing

3. Feature selection

4. Splitting the dataset into training and testing sets

5. Training the machine learning model

6. Evaluating the model performance

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6. Tools & Technologies Used

- Programming Language: Python
- Libraries: NumPy, Pandas, Scikit-learn
- Platform: Jupyter Notebook
- Version Control: GitHub

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7. Machine Learning Models Used

Logistic Regression is used as the primary machine learning model for this project. It is a classification algorithm suitable for binary outcomes such as readmission and no readmission. The model learns the relationship between input features and the target variable to make predictions.

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8. Performance Evaluation

The performance of the model is evaluated using standard metrics such as accuracy, precision, recall, and confusion matrix. These metrics help measure how well the model predicts hospital readmission and identify areas for improvement.

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9. Results and Discussion

The Logistic Regression model successfully predicts hospital readmission with reasonable accuracy. The results show that certain factors such as previous admissions and length of hospital stay play an important role in predicting readmission. The model demonstrates that machine learning can be effectively applied in healthcare analytics.

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10. Conclusion

This project demonstrates the use of machine learning techniques to predict hospital readmission. By analyzing patient data and building a predictive model, hospitals can identify high-risk patients and improve healthcare outcomes. The project highlights the importance of data-driven decision-making in healthcare.

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11. Future Scope

The project can be extended by:

- Using larger and more diverse datasets
- Applying advanced machine learning models
- Integrating real-time hospital data
- Improving prediction accuracy

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12. References

1. Publicly available hospital readmission dataset
2. Scikit-learn Documentation
3. Pandas Documentation